



A Project Synopsis
On
Nagar Sangrah: AI-Enabled Civic Issue Reporting System

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1. ABSTRACT

Urban areas in India face recurring civic challenges such as garbage accumulation, road damage, drainage blockages, and water leakage. Although existing platforms like Swachhata App and FixMyStreet provide mechanisms for reporting civic grievances, they continue to suffer from poor verification, low citizen participation, delayed responses, and limited adoption by government agencies. This project proposes Nagar Sangrah, an AI-enabled, Aadhaar-verified civic issue reporting and tracking platform that ensures report authenticity, location accuracy, government accountability, and transparent resolution workflows. The system integrates AI-based image validation, GPS verification, automated issue routing, role-based dashboards, and a reward-based engagement model to improve the effectiveness of civic governance. Designed for scalability, security, and ease of use, the platform aims to rebuild trust between citizens and municipal bodies while enabling data-driven urban management.

2. PROBLEM STATEMENT

Despite the presence of existing civic issue reporting platforms like **Swachhata App** and **FixMyStreet**, citizens in Jharkhand and many parts of India still experience unresolved urban problems due to low usage, lack of trust, fake/duplicate reports, delayed government action, and no transparent tracking system.

There is no solution that combines:

- Aadhaar-verified citizen identity,
- AI-based media authenticity detection,
- GPS-verified location tagging,
- Automated routing to specific government departments,
- Upvote-based prioritization,
- A reward system for engagement,
- A secure privacy-preserving framework, and
- A seamless mechanism for tracking issue resolution.

Thus, the challenge is to design a scalable, secure, AI-driven civic-tech platform that increases trust, ensures accountability, and enables effective collaboration between citizens and government authorities while addressing issues of misuse, privacy, and adoption.

3. INTRODUCTION

Urban governance in India faces multiple structural challenges. Citizens often resort to phone calls, WhatsApp messages, or in-person visits to municipal offices to report civic issues. This results in:

- Untracked complaints
- Slow response time
- Lack of accountability
- No transparent updates
- Fake or repeated reporting
- Poor prioritization
- Frustration among citizens

Existing platforms fail due to:

1. Low verification of complaints
2. Limited government participation
3. Poor UI/UX and accessibility issues
4. No AI-driven validation or routing
5. No rewards or incentives for users
6. Lack of trust and transparency

Nagar Sangrah introduces an end-to-end reporting system backed by AI validation, Aadhaar-based login, GPS tagging, and a government dashboard to bridge the gap between citizens and civic bodies.

4. LITERATURE REVIEW

Existing Civic Platforms

- Swachhata App: Allows complaint reporting but lacks strong authentication, AI validation, or dashboard transparency. Low adoption by authorities leads to unresolved issues.
- FixMyStreet: Popular in the UK but not widely used in India. Lacks Aadhaar integration and scalable architecture for Indian demographics.
- I Change My City: Encourages citizen reporting but does not solve verification, routing, or government accountability.

Aadhaar Authentication Challenges

Aadhaar APIs are restricted for public developer use. Studies highlight the need for alternate verification methods such as:

- OTP verification using Aadhaar-linked mobile numbers
- DigiLocker-based identity validation
- Government-issued identity proof uploads

AI Image Validation

Research shows CNN-based and EXIF-metadata-based models can detect:

- Edited images
- AI-generated images
- Reposted or old images
- Screenshots vs. real photos

GPS Geolocation Trustworthiness

Studies indicate GPS spoofing can occur; therefore, combining:

- Device GPS
- Network-based location
- EXIF location data
improves accuracy and prevents misuse.

5. WORKFLOW / METHODOLOGY

1. User Login / Verification
 - Aadhaar-linked mobile OTP
 - DigiLocker verification (optional)
 - Role-based login (citizen vs. government staff)
2. Issue Reporting
 - Upload photos/videos
 - Auto-extract metadata (time, location, camera info)
 - AI authenticity check
 - GPS verification
3. Issue Validation
 - Duplicate check
 - Fake image detection
 - Priority assignment based on upvotes and severity
4. Automated Routing
 - Drainage → Water Dept
 - Streetlight → Electricity Dept
 - Roads → Municipal Engineering
 - Garbage → Swachh Bharat Sanitation Unit
5. Government Dashboard Actions
 - Assign staff
 - Update status
 - Upload completion proof
6. Citizen Feedback Loop

- Rate resolution
 - Earn reward points
 - Track historical reports
7. Data Analytics for Governance
- Heatmaps
 - Issue clusters
 - Department performance

6. SYSTEM ARCHITECTURE

Frontend

Using React / Flutter for citizens & staff.

Backend Services

Node.js + Supabase edge functions.

AI Engine

- Fake image detection (CNN model)
- EXIF metadata analyzer
- Duplicate detection using perceptual hashing

Database (Supabase)

For scalable storage, row-level security, and real-time listeners.

Authentication Layer

Aadhaar-linked mobile verification.

Location Engine

GPS + network triangulation + metadata verification.

Government Admin Dashboard

For viewing, assigning, and resolving issues.

Reward Engine

Gamified point and badge system.

7. FUTURE SCOPE

- Integration with municipal ERP systems
- Predictive analytics for identifying high-risk zones
- Voice-based reporting for non-smartphone users
- Regional language support
- Blockchain-based transparency for final reports
- Drone-survey integration for large-scale mapping
- Offline reporting mode for low network areas

10. LIMITATIONS

- Aadhaar authentication APIs are restricted → workaround needed
- AI model accuracy depends on dataset quality
- GPS spoofing still possible though minimized
- Requires government adoption for real impact
- Reward funding may need partnership with CSR/government

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